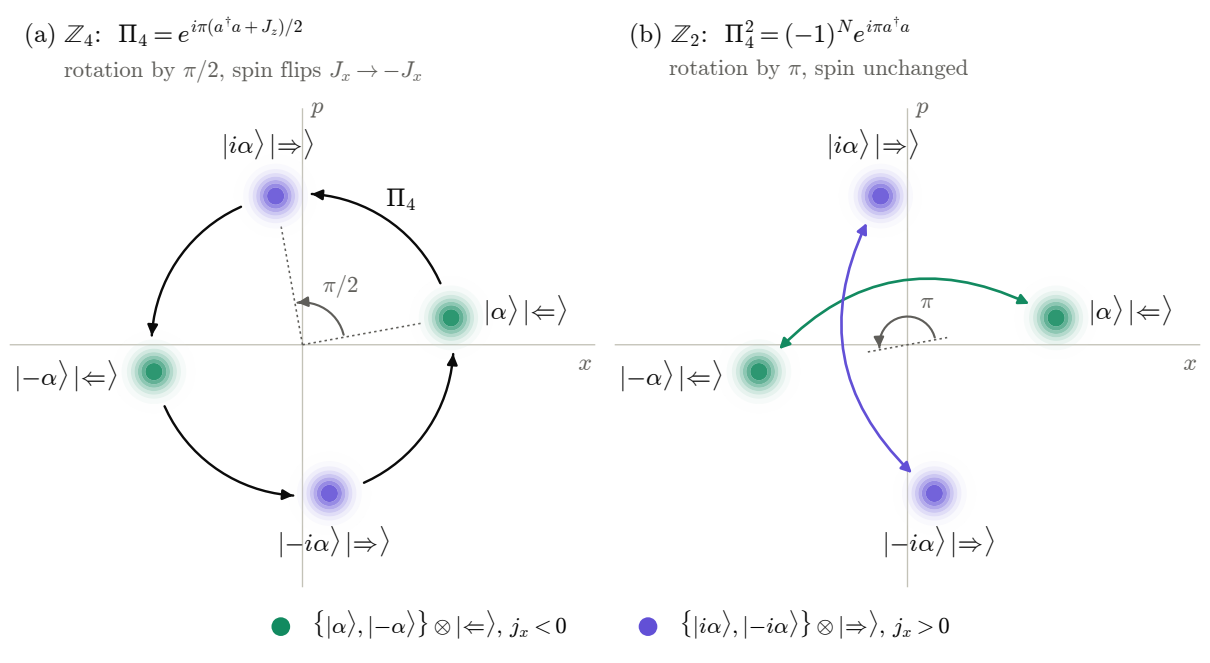

(a) $\mathbb{Z}_4$: $\Pi_4 = e^{i\pi(a^\dagger a + J_z)/2}$
rotation by $\pi/2$, spin flips $J_x \to -J_x$
p
$|i\alpha\rangle\,|\Rightarrow\rangle$
Π_4
$\pi/2$
$|\alpha\rangle\,|\Leftarrow\rangle$
$|-\alpha\rangle\,|\Leftarrow\rangle$
x
$|-i\alpha\rangle\,|\Rightarrow\rangle$
$\{|\alpha\rangle, |-\alpha\rangle\} \otimes |\Leftarrow\rangle$, $j_x < 0$
(b) $\mathbb{Z}_2$: $\Pi_4^2 = (-1)^N e^{i\pi a^\dagger a}$
rotation by π, spin unchanged
p
$|i\alpha\rangle\,|\Rightarrow\rangle$
π
$|\alpha\rangle\,|\Leftarrow\rangle$
$|-\alpha\rangle\,|\Leftarrow\rangle$
x
$|-i\alpha\rangle\,|\Rightarrow\rangle$
$\{|i\alpha\rangle, |-i\alpha\rangle\} \otimes |\Rightarrow\rangle$, $j_x > 0$